Geometry: Pictures, Coordinates, and Curves

Sign up for the Putnam 2026 Competition! (scan the QR code)

Two Circles Meet

Find all intersections of $x^2 + y^2 = 25$ and $(x-4)^2 + y^2 = 9$. More generally, what kind of equation results when the equations of two distinct circles are subtracted?

A Center from Three Points

Find the center of the circle through $(0,0)$, $(4,0)$, and $(1,3)$. If three noncollinear points have rational coordinates, must the center of their circle have rational coordinates?

A Shear and a Triangle

The map $T(x,y) = (2x+y, x+y)$ sends the triangle with vertices $(0,0), (1,0), (0,1)$ to another triangle. Find both areas. For a general linear map $T(x,y) = (ax+by, cx+dy)$, by what factor are triangle areas multiplied? Include the case $ad - bc = 0$.

A Visit to the River

A traveler goes from $P = (1,2)$ to a point X on the x-axis, then to $Q = (5,3)$. Find the shortest possible journey and the point X where it is attained.

A Chord above a Curve

Let $A = (1,1)$ and $B = (4, 1/4)$ on $y = 1/x$. Find the area bounded by the chord AB and the intervening hyperbola arc. For $P = (t, 1/t)$ with $1 < t < 4$, express the area of triangle APB as a function of t.

Rational Points on a Circle

What is the maximum number of points with rational coordinates that can lie on a circle whose center does not have rational coordinates?

Source: Putnam 2008 B1

A Triangle in a Wedge

Given $P = (a, b)$ with $0 < b < a$, determine the minimum perimeter of a triangle with one vertex P, one vertex on the x-axis, and one vertex on $y = x$. You may assume a minimizing triangle exists.

Source: Putnam 1998 B2

Two Hyperbolic Segments

Let A, B lie on the same branch of $xy = 1$. A point P between them on the hyperbola maximizes the area of triangle APB. Show that the region bounded by chord AP and the hyperbola has the same area as the region bounded by chord PB and the hyperbola.

Source: Putnam 2015 A1